

Anthony Okolie

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SUMMARY

Computer Science graduate with experience building full-stack and backend-heavy applications using TypeScript, React, Node.js, Java, PostgreSQL, and cloud technologies. Built distributed real-time systems, AI-powered data pipelines, and scalable backend services involving REST APIs, authentication, WebSockets, Redis, and database design. Interested in building high-impact software systems that combine modern backend infrastructure, AI capabilities, and strong user experiences.

EDUCATION

University of Manitoba

Sep 2021 - Jun 2026

Bachelor of Computer Science

- **Achievements:** Graduated from the Honours program, Deans Honors List (2021)
- **Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering, OOP, Data Mining

SKILLS

- **Languages:** Go, Java, TypeScript, JavaScript, Python
- **Frontend:** React, Angular, HTML5, CSS3
- **Backend:** Node.js, Express, REST APIs, WebSockets, Spring Boot, JWT Authentication, Redis
- **Cloud & Tools:** Docker, CI/CD, Azure(basic), Git, GitHub, AWS (Elastic Beanstalk, RDS), Excel(data analysis, structured datasets)
- **Databases & ORM:** PostgreSQL, SQL, MongoDB, Prisma
- **Testing:** JUnit, Vitest, Cypress
- **Concepts:** Distributed Systems, Real-Time Systems, Database Design, Authentication, Concurrent Systems, API Design, Data Pipelines

PROJECTS

Real-Time Collaboration Board

Node.js, Express, React, TypeScript, PostgreSQL, Redis, Prisma, AWS, Docker

GitHub: <https://github.com/AnOkolie/collabBoard>

- Built a real-time collaborative editing platform enabling multiple users to edit shared content simultaneously.
- Implemented WebSocket-based state synchronization to maintain consistency across concurrent clients.
- Integrated REST APIs with PostgreSQL for persistent data storage.
- Implemented authentication and session management using JWT to secure collaborative workspaces.
- Deployed application using Docker with AWS Elastic Beanstalk and RDS for scalable backend hosting; supported testing with 30 beta users.
- Evaluated trade-offs between real-time synchronization complexity and system performance, iterating on architecture decisions to balance responsiveness and reliability.

Secure Secret Vault API

Go, PostgreSQL, Docker, Rest APIs

GitHub: <https://github.com/AnOkolie/secret-vault>

- Built a secure backend system in Go for encrypted storage and retrieval of sensitive user data via authenticated REST APIs.
- Implemented JWT-based authentication, role-based access control, and request validation for secure data operations.
- Designed PostgreSQL schema supporting data lifecycle management and audit logging for secure system traceability.
- Developed encryption workflows for protecting sensitive data at rest and in transit.
- Containerized application using Docker and implemented automated testing workflows for reliability and maintainability.
- Added structured logging and error handling to support production-grade debugging and observability.

News Intelligence Pipeline

JavaScript, Playwright, OpenAI, PostgreSQL, Vitest

GitHub: <https://github.com/AnOkolie/news-scrapper>

- Built an AI-powered pipeline to scrape, classify, and summarize Hacker News content using Playwright, OpenAI, and PostgreSQL.
- Automated ingestion and processing of 100+ articles per run, generating structured metadata including sentiment, topics, and keyword analysis.
- Designed PostgreSQL-backed workflows for storing, querying, and analyzing processed article data.
- Developed a CLI analytics dashboard surfacing trending topics, sentiment breakdowns, and feed-level insights.
- Developed automated testing workflows using Vitest and Playwright to improve pipeline reliability and validation accuracy.

EXPERIENCE

DNOJ Accounting and Tax

May 2025 - Jul 2025

Web Designer/Data Assistant

- Worked with structured financial data to support reporting and dashboard deployments.
- Cleaned and validated datasets to ensure accuracy for internal reporting systems
- Documented data workflows and collaborated with stakeholders to improve data usability.
- Communicated technical findings and insights to non-technical users.

Canada Revenue Agency

May 2022 - Dec 2022

Tax Processing Assistant (T1 Matching Assessor)

- Processed 1,000+ tax cases with high accuracy in a financial services environment.
- Communicated with clients to request documentation and ensure compliance requirements were met.
- Identified and resolved data inconsistencies using structured validation procedures.